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HBASE

A Distributed NoSQL Database for Scalable Storage

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INTRODUCTION

What is Apache HBase?

Apache HBase is a distributed, column-oriented NoSQL database built to store and process massive datasets across clusters.

- Distributed and horizontally scalable
- Column-family data model
- Real-time random read/write access
- Fault tolerant via distributed storage
- Schema flexible

Why Was HBase Created?

Limitations of Traditional Databases :

- Difficulty handling massive datasets
- Expensive vertical scaling
- Rigid schemas
- Reduced performance at scale

The Need :

Organizations required systems that could:

- Scale across machines
- Handle billions of rows
- Support high write throughput
- Provide fault tolerance

How HBase Works

High-Level Working :

- Client sends read/write request
- HBase Master manages schema and region allocation
- RegionServers store and process data
- Data is persisted in Hadoop Distributed File System (HDFS)

Result:

- Automatic data distribution
- High availability
- Scalable performance

Distributed System Principles Used in HBase

- Scalability
- Fault tolerance
- Strong consistency
- Partitioning and Parallelism
- High availability

HBase Data Model

- A column-family–oriented data model used to store large-scale distributed data.
- Organizes data into tables, rows, column families, and columns.
- Each row is identified by a unique row key for fast access.
- Supports dynamic columns, allowing schema flexibility.
- Stores data as key–value pairs within cells.
- Provides versioning, enabling multiple timestamped values per cell.

HBase Data Model: Logical View & Components

Row Key: Unique identile identifier, sorted

Column Family: Defined at table creation

Column Qualifier) Added on the fly

Roow Keey

Row Key	Column Personal	Professional
User_001	Name: "Alice"	Role: "Engineer
Age: "30	Name: "30	Dept: "DevOPs"
User_002	Bob	Role: "Manager"

Cell: Row Key + CF CF + CQ + Value

Row Key + CF + CQ + Value

Timesttop/Version: Multiple versions per cell

Tables > Row Key > Column Families > Column Qualifiers (Columns) > Cells > Timestapm

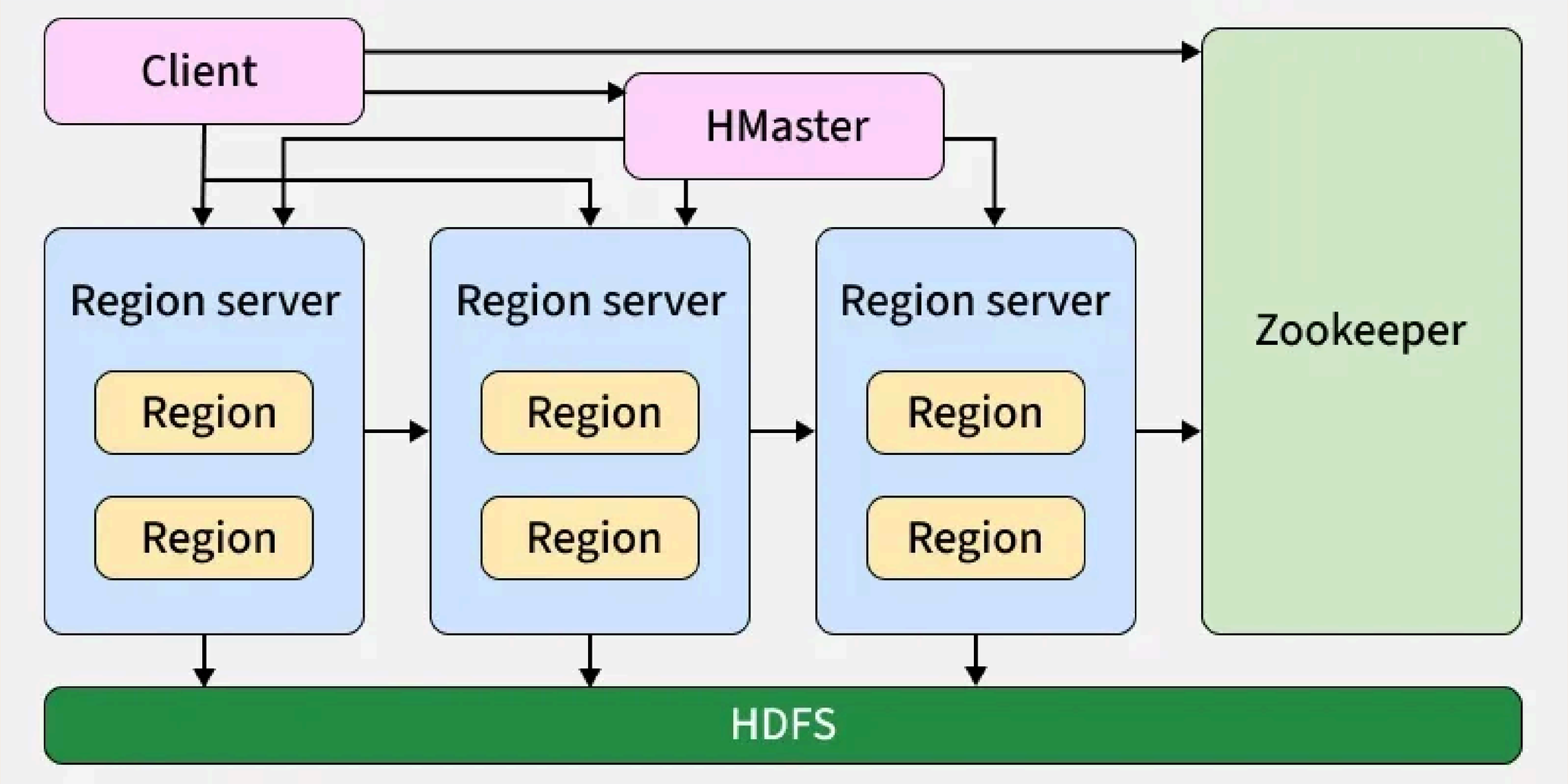
Strengths of the HBase Model

- Handles massive datasets efficiently
- Supports high write throughput
- Allows flexible column definitions
- Optimized for sparse data
- Built-in version control
- Designed for horizontal scalability

Limitations

- Understanding constraints demonstrates deeper technical awareness.
- No native JOIN capability
- Limited support for complex transactions
- Requires careful row-key design
- Not suited for heavy relational querying
- HBase is therefore best viewed as a distributed storage engine rather than a full relational replacement.

HBASE ARCHITECTURE



HMASTER - The main coordinator of the hbase cluster

- Assigns regions (Data chunks) to region servers
- Manages table operations like create, delete, and modify

REGION SERVERS - Divided horizontally into smaller parts called regions

- Store and manage regions, each containing data
- Handles read and write requests from clients.
- Used for storing recently written data

ZOOKEEPER-works like a traffic controller for HBase

- Helps client find which region server holds which data
- Maintains cluster configurations

HDFS -Where the data is actually stored in HBase

- NameNode → manages metadata
- DataNodes → store actual data blocks
- HBase stores:

  -HFiles

  -Write Ahead Logs (WAL)

Problem Statement

City Environmental Data Management Challenge

Smart City Environmental Data Management using HBase

Existing Problems:

- Massive Data Volume
- Real Time Quering Problems
- Scalability Issues.

Why Apache HBase?

- Handles massive data
- Fast real time access
- Horizontal Scalability.